Week-3 DL

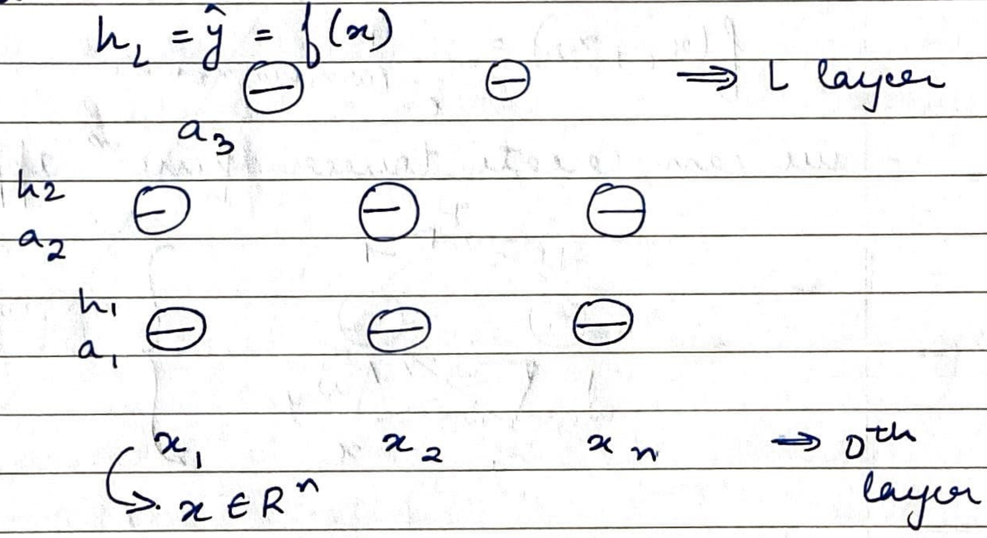
L1 FFNN (Feed Forward Neural Network)

- input has n-dim. vector

network with (L(-1)) hidden layers with n neurons.

1 output layer with k neurons (k classes)

each neuron has 2 parts — pre-activaⁿ & activaⁿ



$W_i \in R^{n \times n}$ & $b_i \in R^n$ are wts. & bias btw. layers $i-1$ & i $(0 < i < L)$

$W_L \in R^{k \times n}$ & $b_L \in R^k$ btw. final hidden layer & output layer

pre-activaⁿ ← $\underbrace{a_i(x)}_{R^n} = \underbrace{b_i}_{R^n} + \underbrace{W_i}_{R^{n\times n}} \underbrace{h_{i-1}(x)}_{R^n}$

activaⁿ ← $h_i(x) = g(a_i(x))$

↳ activaⁿ funcⁿ (log, tanh, etc)

all are working element-wise

actuia^n at output layer $\leftarrow$ $f(x) = h_L(x) = O(a_L(x))$

- we have our typical ML setup

$\{x_i, y_i\}_{i=1}^{N}$

model $\swarrow$ $\hat{y}_i = f(x_i) = O(W_3\, g(W_2\, g(W_1 x_i + b_1) + b_2) + b_3)$

$\theta = W_1, \dots W_L, b_1 \dots b_L$

GD with backprop.

Obj. $\min \frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{k} (\hat{y}_{ij} - y_{ij})^2$

$\min. L(\theta) \Rightarrow$ same func^n of params!

L2 Learning Params

- Recall GD

$\theta_{t+1} \leftarrow \theta_t - \eta \nabla \theta_t$

$\nabla \theta_t = \left[\frac{\partial L(\theta)}{\partial W_t}, \frac{\partial L(\theta)}{\partial b_t} \right]^T$

in FFNN, $\theta = W_1 \dots W_L, b_1 \dots b_L$

- now,

$\nabla \theta$ has

$\nabla W_1 \text{ ---- } \nabla W_{L-1} \in \mathbb{R}^{n \times n}, \quad \nabla W_L$

$\nabla b_1 \text{ --- } \nabla b_{L-1} \in \mathbb{R}^{n}, \quad \nabla b_L$

- now how to choose $L(\theta)$? , how to compute $\nabla\theta$.

L3 Output & Loss Func^n

- choice of loss func^n depends on the problem. majorly we have 2 problems — classifica^n & regression
- loss func^n should capture $\hat{y}_i$ deviates from y.

MSE for Reg. tasks $L(\theta) = \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{m}(\hat{y}_{ij} - y_{ij})^2$

- what should be my 'O' if $y \in R$? best to have a linear func^n.

$$f(x) = h_L = O(a_L)$$
$$= W_O a_L + b_O$$

in Regression

- for classifica^n, y is probab. distribu^n

$$a_L = W_L h_{L-1} + b_L$$

$$\hat{y}_j = O(a_L)_j = \frac{e^{a_{Lj}}}{\sum_{i=1}^{k} e^{a_{Li}}}$$

softmax func^n

j^{th} element of $\hat{y}$

j^{th} element of the vector a_L

exp. $\Rightarrow$ all -ve values becomes positive

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- Cross entropy

$$L(\theta) = -\sum_{c=1}^{k} y_c \log \hat{y}_c$$

$y_c = 1$ if $c = l$ (true class)
$\quad = 0$ o.w.

$$\therefore L(\theta) = -\log \hat{y}_l$$

log-likelihood $\qquad$ predicted probability of the true class.

$$\min_{\theta} L(\theta) = -\log \hat{y}_l$$

& yes $\hat{y}_l$ is func^n of $\theta \in [W_1, \ldots, W_L, b_1, \ldots, b_L]$

		Real output	probab.
Output	$\rightarrow$	Linear	softmax
Loss	$\rightarrow$	MSE	CE
		(Regression)	(Classifica^n)

L4 **Backpropaga^n**

$$x_1 \xrightarrow[W_{111}]{} \underset{a_{11}\ h_{11}}{\bigcirc} \xrightarrow{W_{211}} \underset{a_{21}\ h_{21}}{\bigcirc} \cdots \xrightarrow[W_{L11}]{} \underset{a_{L1}}{\bigcirc} \rightarrow \hat{y}$$

$$\frac{\partial L(\theta)}{\partial W_{111}} = \frac{\partial L(\theta)}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial a_{L11}} \frac{\partial a_{L11}}{\partial h_{21}} \frac{\partial h_{21}}{\partial a_{21}} \cdots$$

if I know this much, then this can be reused again.

$$\frac{\partial L(\theta)}{\partial W_{111}} = \overbrace{\frac{\partial L(\theta)}{\partial h_{11}}}\ \frac{\partial h_{11}}{\partial W_{111}}$$

- we get a particular loss at the output & we try to fig. out who is responsible for this loss. ⇒ depends on wts. & bias.
- we will find gradient wrt output units, hidden units & finally wts. & bias.

L5 <u>Gradient wrt Output</u>

$$\frac{\partial L(\theta)}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial a_3}$$

$\nabla_{a_L} L(\theta) \longrightarrow$ our interest

$$L(\theta) = -\log \hat{y}_l$$

$$\frac{\partial L(\theta)}{\partial \hat{y}_i} = \frac{\partial(-\log \hat{y}_l)}{\partial \hat{y}_i}$$

$$= -\frac{1}{\hat{y}_l} \quad \text{if } i = l$$

$$= 0 \quad \text{o.w.}$$

so, $$\frac{\partial L(\theta)}{\partial \hat{y}_i} = -\frac{1_{i=l}}{\hat{y}_l}$$

$$\nabla_{\hat{y}} L(\theta) = \begin{bmatrix} \frac{\partial L(\theta)}{\partial \hat{y}_1} \\ \vdots \\ \frac{\partial L(\theta)}{\partial \hat{y}_k} \end{bmatrix} = -\frac{1}{\hat{y}_l} \begin{bmatrix} 1_{l=1} \\ 1_{l=2} \\ \vdots \\ 1_{l=k} \end{bmatrix}$$

$$= -\frac{1}{\hat{y}_l} e_l$$

$$\frac{\partial L(\theta)}{\partial a_{Li}} = \frac{\partial(-\log \hat{y}_\ell)}{\partial a_{Li}}$$

$$= \frac{\partial(L(\theta))}{\partial \hat{y}_\ell} \frac{\partial \hat{y}_\ell}{\partial a_{Li}}$$

$$\frac{\partial L(\theta)}{\partial a_{Li}} = \frac{\partial}{\partial a_{Li}} - \log \hat{y}_\ell = \frac{-1}{\hat{y}_\ell} \frac{\partial}{\partial a_{Li}} \hat{y}_\ell$$

$$= \frac{-1}{\hat{y}_\ell} \frac{\partial}{\partial a_{Li}} \text{softmax}(a_L)_\ell$$

$$= \frac{-1}{\hat{y}_\ell} \frac{\partial}{\partial a_L} \frac{e^{(a_L)_\ell}}{\sum e^{(a_L)_\ell}}$$

$$= \frac{-1}{\hat{y}_\ell} (1_{(\ell=i)} \hat{y}_\ell - \hat{y}_\ell \hat{y}_i)$$

$$= -(1_{(\ell=i)} - \hat{y}_i)$$

$$\nabla_{a_L} L(\theta) = \begin{bmatrix} \frac{\partial L(\theta)}{\partial a_{L1}} \\ \vdots \\ \frac{\partial L(\theta)}{\partial a_{Lk}} \end{bmatrix} = \begin{bmatrix} -1_{\ell=1} - \hat{y}_1 \\ \vdots \\ -1_{\ell=k} - \hat{y}_k \end{bmatrix}$$

$$= -(e(\ell) - \hat{y})$$

$\underline{\text{Gradient wrt. hidden units}}$

$$\underbrace{\frac{\partial a_3 h_2}{\partial h_2 \partial a_2} \quad \frac{\partial a_2}{\partial h_1} \frac{\partial h_1}{\partial a_1}}_{\text{hidden layers}}$$

chain rule along multiple paths $\left\{ \frac{\partial p(z)}{\partial z} = \sum_m \frac{\partial p(z)}{\partial q_m} \frac{\partial q_m}{\partial z} \right.$

$$\left.\begin{aligned} p(z) &\Rightarrow L(\theta) \\ z &\Rightarrow h_{ij} \\ q_m(z) &\Rightarrow a_{Lm} \end{aligned}\right\} \text{ in our case}$$

$$\frac{\partial L(\theta)}{\partial h_{ij}} = \sum_{m=1}^{k} \frac{\partial L(\theta)}{\partial a_{i+1,m}} \frac{\partial a_{i+1,m}}{\partial h_{ij}}$$

$$= \sum_{m=1}^{k} \frac{\partial L(\theta)}{\partial a_{i+1,m}} W_{i+1,m,j}$$

$$\boxed{a_{i+1} = W_{i+1} h_{ij} + b_{i+1}}$$

$$\nabla_{a_{i+1}} L(\theta) = \begin{bmatrix} \frac{\partial L(\theta)}{\partial a_{i+1,1}} \\ \vdots \\ \end{bmatrix} ; \quad W_{i+1,\cdot,j}$$

$$= \begin{bmatrix} W_{i+1,1,j} \\ \vdots \\ W_{i+1,k,j} \end{bmatrix}$$

$$(W_{i+1,\cdot,j})^T \nabla_{a_{i+1}} L(\theta) = \sum_{m=1}^{k} \frac{\partial L(\theta)}{\partial a_{i+1,m}} W_{i+1,m,j}$$

$$\nabla_{h_i} L(\theta) = \begin{bmatrix} \frac{\partial L(\theta)}{\partial h_{i1}} \\ \vdots \\ \frac{\partial L(\theta)}{\partial h_{in}} \end{bmatrix} = \begin{bmatrix} (W_{i+1,\cdot,1})^T \nabla_{a_{i+1}} L(\theta) \\ \vdots \\ (W_{i+1,\cdot,n})^T \nabla_{a_{i+1}} L(\theta) \end{bmatrix}$$

$$= (W_{i+1})^T (\nabla_{a_{i+1}} L(\theta))$$

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$$\frac{\partial L(\theta)}{\partial a_{ij}} = \frac{\partial L(\theta)}{\partial h_{ij}} \frac{\partial h_{ij}}{\partial a_{ij}}$$

$$= \frac{\partial L(\theta)}{\partial h_{ij}} g'(a_{ij})$$

$$\nabla a_i L(\theta) = \begin{bmatrix} \frac{\partial L(\theta)}{\partial h_{ij}} g'(a_{i1}) \\ \vdots \\ \frac{\partial L(\theta)}{\partial h_{ij}} g'(a_{in}) \end{bmatrix}$$

$$= \nabla h_i L(\theta) \odot [\ldots, g'(a_{ik}), \ldots]$$

L7 Gradient wrt Params

$$\left.\frac{\partial a_1}{\partial W_{111}}\right\}$$ on the final wts.

$$a_k = b_k + W_k h_{k-1}$$

$$\frac{\partial L(\theta)}{\partial W_{kij}} = \frac{\partial L(\theta)}{\partial a_{ki}} \frac{\partial a_{ki}}{\partial W_{kij}}$$

$$= \frac{\partial L(\theta)}{\partial a_{ki}} h_{k-1,j}$$

$$\nabla W_k L(\theta) = \begin{bmatrix} \frac{\partial L(\theta)}{\partial W_{k11}} & \cdots & \\ \vdots & & \frac{\partial L(\theta)}{\partial W_{k33}} \end{bmatrix} \quad \frac{\partial L(\theta)}{\partial W_{kij}}$$

$$= \nabla a_k L(\theta) \cdot h_{k-1}^T$$

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$$\frac{\partial L(\theta)}{\partial b_{ki}} = \frac{\partial L(\theta)}{\partial a_{ki}} \frac{\partial a_{ki}}{\partial b_{ki}}$$

$$= \frac{\partial L(\theta)}{\partial a_{ki}}$$

$$\nabla b_k L(\theta) = \nabla a_k L(\theta)$$

L8 Pseudocode

$h_1 \dots h_{L-1}, a_1 \dots a_L, \hat{y} \Rightarrow$ forward pass

$\nabla\theta_t = \text{backward}(h_1 \dots h_{L-1}, a_1 \dots a_L, y, \hat{y})$

$\theta_{t+1} \leftarrow \theta_t - \eta \nabla\theta_t$

- Forward

$$\left.\begin{aligned} a_k &= b_k + W_k h_{k-1} \\ h_k &= g(a_k) \end{aligned}\right\} L-1 \text{ times}$$

$$\left.\begin{aligned} a_L &= b_L + W_L h_{L-1} \\ \hat{y} &= O(a_L) \end{aligned}\right\} L^{th} \text{ layer}$$

- Backward

$\nabla a_L L(\theta) = -(e(y) - \hat{y}) \Rightarrow$ output

from L to 1.

$$\begin{cases} \nabla_{W_k} L(\theta) = \nabla_{a_k} L(\theta)\, h_{k-1}^T & \Rightarrow \text{params} \\ \nabla_{b_k} L(\theta) = \nabla_{a_k} L(\theta) & \\ \nabla_{h_{k-1}} L(\theta) = W_k^T \nabla_{a_k} L(\theta) & \Rightarrow \text{layer} \\ \nabla_{a_{k-1}} L(\theta) = \nabla_{h_{k-1}} L(\theta) \odot [\dots, g'(a_{k-1,j}) \dots] & \end{cases}$$

$\hookrightarrow$ pre-activation

$- \quad \sigma(z) = \frac{1}{1+e^{-z}} \qquad g'(z) = \sigma(z)(1-\sigma(z))$

$\tanh(z) = \frac{e^{z}-e^{-z}}{e^{z}+e^{-z}} \qquad g'(z) = 1-(g(z))^2$